

ELT3207 Meas. & Digital Control Basics

Lecture 7: Uncertainty analysis

Hoang Gia Hung

Faculty of Electronics and Telecommunications
University of Engineering and Technology, VNU Hanoi

Recap & Agenda

- ❑ Random error analysis
 - General probabilistic concepts
 - Steps to perform statistical inferences
- ❑ Parameter estimation
 - Interval estimation of population mean: Central limit theorem & Student's t distribution approximation.
 - Interval Estimation of the Population Variance: Chi-squared distribution
- ❑ Statistical methods for rejecting questionable data points
 - Static measurements: three-sigma test, modified Thompson π technique.
 - x-y Data Sets: standardized residual.
- ❑ **Today's lecture:** Uncertainty analysis

What is Uncertainty?

- ❑ The gap: error = how close the measurement is to the true value?
 - The true value of a physical quantity is generally unknown, so exact error cannot be calculated.
 - Can only estimate a *range of probable error* in the result of a measurement → the **uncertainty** in the measured value.
 - There are several sources of uncertainty (random, systematic, ...)
- ❑ The process of identifying all probable sources of uncertainties & estimating the uncertainty for the final results of a measurement is called **uncertainty analysis**.
 - In lecture 1, we know how to combine different types **errors** to determine the final error, will it be the same to combine **uncertainties** from different sources?
 - **Go back**: This analysis assists in selecting measurement techniques and devices during the **design stage**.

Categorizing Measurement Errors

❑ Illegitimate Errors

- Blunders result from mistakes in procedure → we must be careful.
- Computational or calculation errors after the experiment.

❑ Systematic or Bias Errors

- An error that persists and tends to stay constant from trial to trial
 - Systematic errors/faulty equipment can be corrected through calibration
 - Consistent or recurring human errors: observer bias
- Can be determined via comparison/alternate measurements.

❑ Random or Precision Errors:

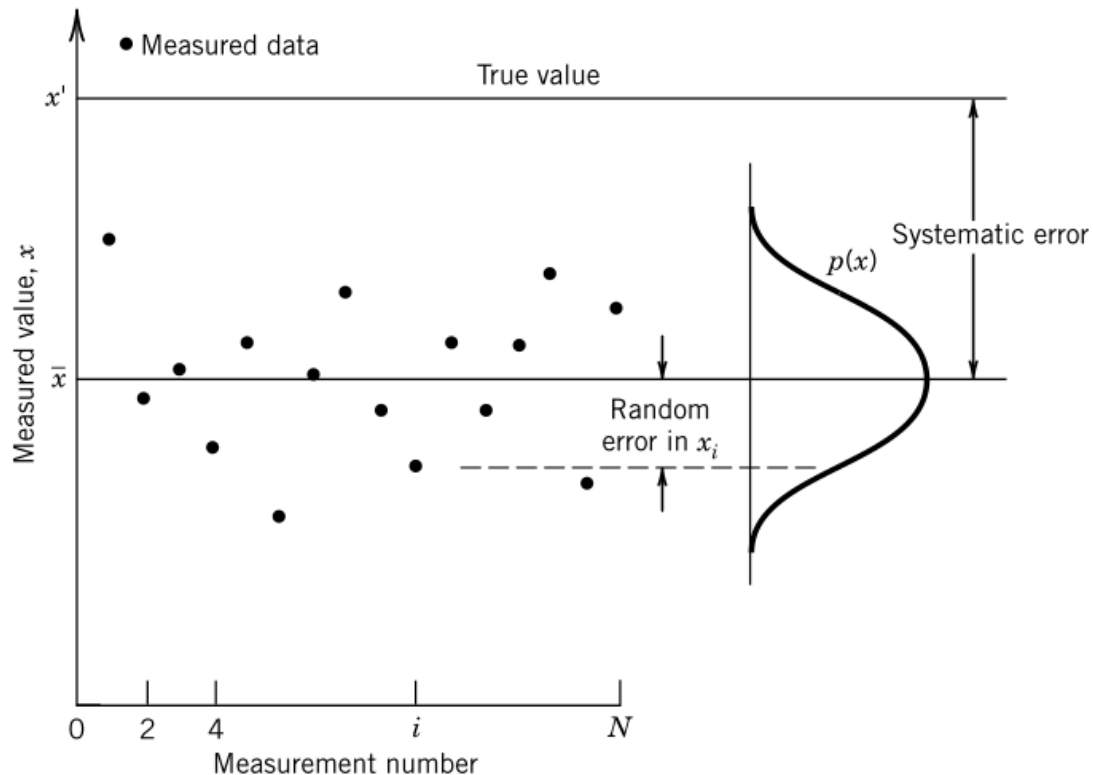
- The deviation resulting from the finite precision of the measurement method, such as instrument friction or hysteresis, calibration drift, test condition variations or environmental effects.
- Reduce random errors by conducting more experiments/take more data.
- Because of the varying nature of random errors, exact values cannot be known while probable estimates of the error can be made through statistical analyses.

The Two Tracks (Systematic vs. Random)

- ❑ In uncertainty analyses, we normally keep separate track of the systematic or bias uncertainty (B) and the random uncertainty (P).
 - A systematic error does not vary during experiments and is independent of the sample size, whereas, the random uncertainty depends on the sample size → it's desirable to handle them separately in the uncertainty analysis.
 - Random uncertainty has a well-established statistical basis while systematic uncertainty does not lend itself to rigorous statistical analysis.
- ❑ Uncertainty = range of **probable** error → uncertainty usually associates with a **level of confidence** (expressed in %).
 - A random uncertainty with confidence level of 95% implies that the actual error will be less than the estimate for 95% of long-run repeated measurements.
- ❑ The systematic and random uncertainties need to be evaluated with the same confidence level to be combined (usually 95%).

Visualizing the Components

- Relationship between the true value, systematic, and random errors on repeated measurements.



Random uncertainty

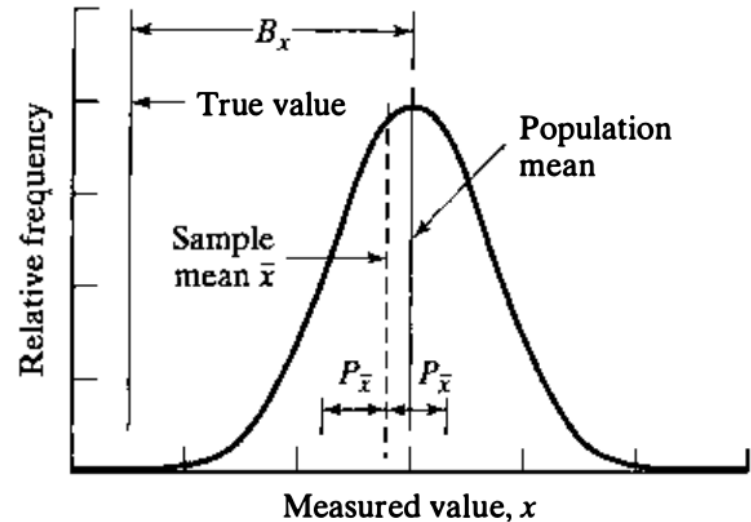
- ❑ If the variable x is measured n times, then

- the mean of the **sample** is

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$$

- the standard deviation of the **sample** is

$$S_x = \left(\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} \right)^{1/2}$$



- ❑ The mean of a batch of measurements (sample mean) differs from the population mean and has an uncertainty interval of $\pm P_{\bar{x}}$.
- For large n , $\bar{x}$ follows Gaussian distribution $\rightarrow P_{\bar{x}} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \approx z_{\alpha/2} \frac{S_x}{\sqrt{n}}$ with confidence level $1 - \alpha$.
- For small n , $\bar{x}$ follows Student's t-distribution $\rightarrow P_{\bar{x}} = \pm t \frac{S_{\bar{x}}}{\sqrt{n}}$ where the value of t is obtained from the t-distribution table for a given confidence level as well as the degree of freedom $\nu = n - 1$.

Handling Small-Sized Samples

- ❑ Usually, S_x is determined in tests with a large sample size ($n > 30$). In addition, a confidence level of 95% is commonly used to determine the random uncertainty interval.
 - For these two conditions, the value of t is 2.0
- ❑ In some cases, the input elemental random uncertainties are based on different samples with different sample sizes.
 - If any of the samples has a size less than 30, it is necessary to find a combined value of ν , as

$$\nu = \frac{\sum_{i=1}^m (S_i^2)^2}{\sum_{i=1}^m (S_i^4 / \nu_i)}$$

where ν_i is the degree of freedom of the individual elemental error (number of data values in the sample, minus 1), m is the number of elemental random uncertainties.

Systematic uncertainty

❑ Calibration

- While the process, of course, is performed to minimize resulting systematic errors, some residual error will remain, resulting in calibration uncertainties. The causes include factors such as uncertainty in standards, uncertainty in the calibration process, and randomness in the process.

❑ Data Acquisition

- Errors in data acquisition equipment such as measurand loading, A/D conversion uncertainties, and uncertainties in recording or indicating devices

❑ Data Reduction

- Caused by a variety of errors and approximations that are used in computers and calculators such as interpolating, curve fitting, and differentiating data curves.

❑ Errors of Method

- Personal errors/blunders

Systematic Uncertainty - Calibration

- ❑ Calibration uncertainties are uncertainties that originate in the calibration process.
- ❑ Calibration minimizes resulting systematic errors, some residual errors remain, resulting in calibration uncertainty.
- ❑ Calibration errors tend to enter through these principal sources:
 - Uncertainty in calibration process.
 - Randomness in the calibration process.
 - Uncertainty in standards.

Systematic Uncertainty - DAQ

- ❑ All uncertainties, that arise when the actual measurements are made, are referred to as data acquisition uncertainties.
- ❑ Some common data acquisition errors are listed below.

Table 5.2 Data-Acquisition Error Source Group

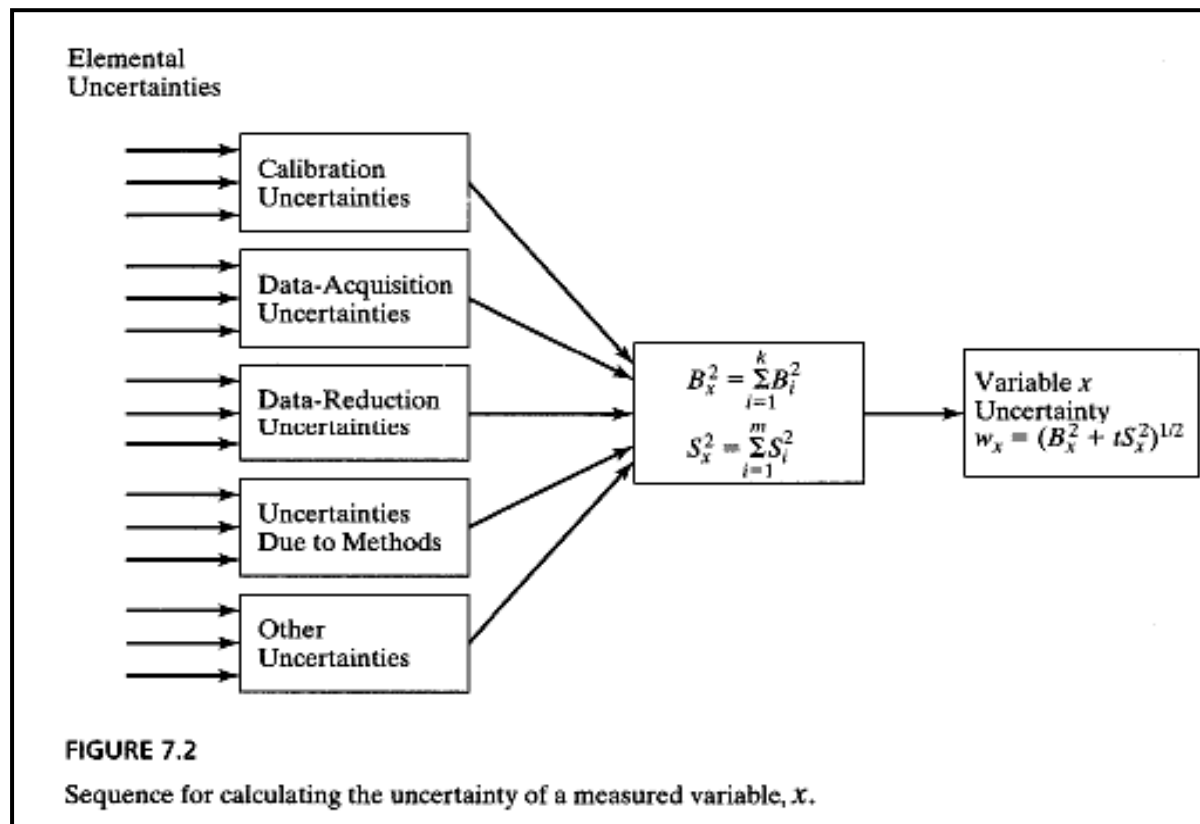
Element	Error Source ^a
1	Measurement system operating conditions
2	Sensor-transducer stage (instrument error)
3	Signal conditioning stage (instrument error)
4	Output stage (instrument error)
5	Process operating conditions
6	Sensor installation effects
7	Environmental effects
8	Spatial variation error
9	Temporal variation error
etc.	

Systematic Uncertainty – Data Reduction and Methods

- ❑ **Data-reduction uncertainties** are caused by a variety of errors & approximations that are used in the data-reduction process
 - E.g., interpolation, curve fitting, and correlation.
 - The rounding-off of the resulted values also introduces the error called truncation error.
- ❑ **Uncertainties due to methods** are uncertainties originate from techniques or methods inherent in the measurement process
 - E.g., assumptions or constants in the calculation routines, disturbance effects caused by instrumentation, spatial effects, and uncertainties due to instability, non-repeatability, and hysteresis of test processes.
- ❑ The categories are primarily a bookkeeping and diagnostic tool.
 - Since elemental uncertainties are combined by using the square root of the sum of the squares (RSS), it does not matter into which category an elemental error is placed.

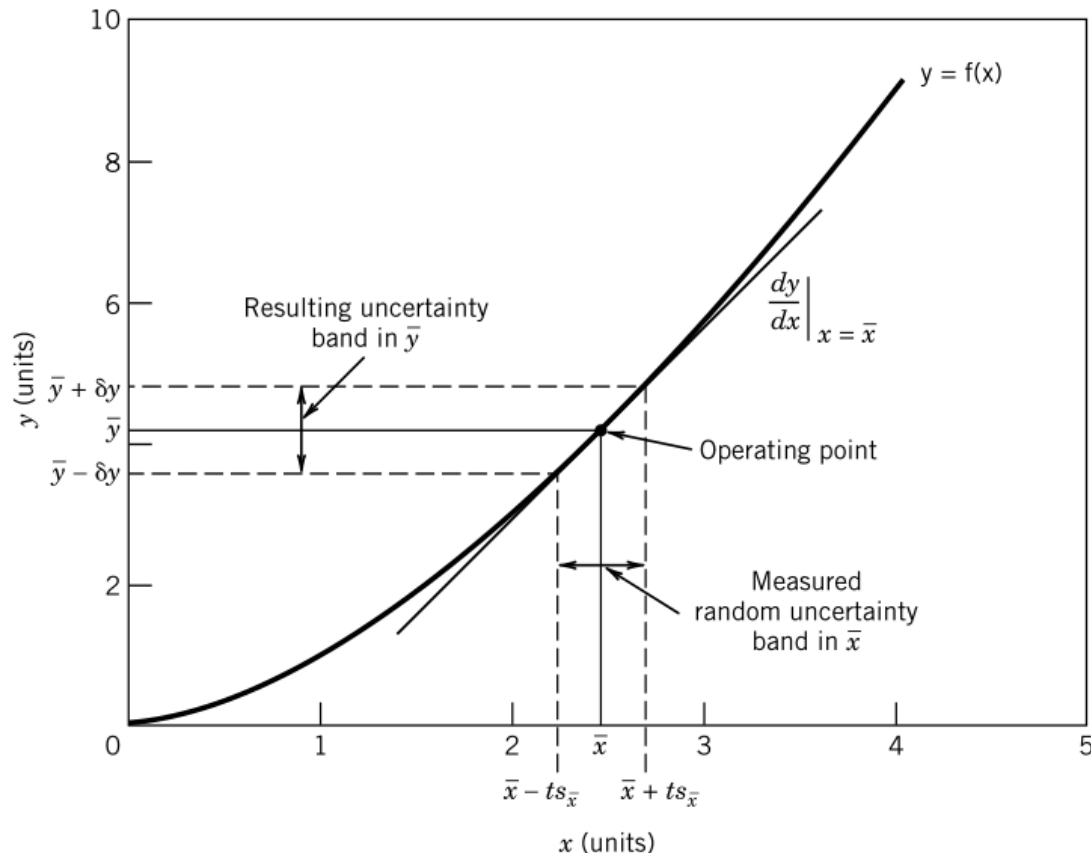
Combining Elemental Uncertainties

- ❑ To obtain the total uncertainty of a measurement, elemental systematic and random uncertainties must be **combined**.
 - The combination of systematic uncertainties and the standard deviations are illustrated as follows.



Propagation of uncertainties (1)

- Suppose we have measured variables V and I with corresponding uncertainties. If we compute $P = V/I$, the uncertainties in V and I propagate into the uncertainty of the result, P .



Propagation of Uncertainties (2)

- In general, consider $R = f(x_1, x_2, \dots, x_n)$. The propagation of uncertainties in measurands x_i 's into the uncertainty of R is

$$\delta R = \delta x_1 \frac{\partial R}{\partial x_1} + \delta x_2 \frac{\partial R}{\partial x_2} + \dots + \delta x_n \frac{\partial R}{\partial x_n} = \sum_{i=1}^n \delta x_i \frac{\partial R}{\partial x_i} \quad (1)$$

- δx_i 's are the uncertainties in the variable (usually denoted by w_{x_i}) while δR is the uncertainty in the result (usually denoted by w_R). All uncertainty estimates should be made to the **same confidence level**.
- $\frac{\partial R}{\partial x_i}$ is called the **sensitivity coefficient** of result R with respect to variable x_i .

- If individual variable uncertainties cancel out each other $w_R = 0 \rightarrow$ Eq. (1) will not produce a realistic value for w_R . Better estimate:

- **Maximum uncertainty**: $w_R = \sum_{i=1}^n \left| w_{x_i} \frac{\partial R}{\partial x_i} \right| \quad (2)$

- (2) usually produces an unreasonably high estimate for w_R .

- The **root of the sum of the squares** : $w_R = \left(\sum_{i=1}^n \left[w_{x_i} \frac{\partial R}{\partial x_i} \right]^2 \right)^{1/2} \quad (3)$

- The confidence level in the uncertainty in the result, R , will be the same as the confidence levels of the uncertainties in the x_i 's.
- Each of the measured variables should be **independent** of each other.

Propagation of Uncertainties (2)

- ❑ If the result R is dependent only on the product of the measured variables, i.e. $R = C x_1^a x_1^b \dots x_n^N$, (3) will take a simpler form

$$\frac{w_R}{R} = \left\{ \left(a \frac{w_{x_1}}{x_1} \right)^2 + \left(b \frac{w_{x_2}}{x_2} \right)^2 + \dots + \left(N \frac{w_{x_n}}{x_n} \right)^2 \right\}^{1/2}.$$

- ❑ **Example:** To calculate the power consumption in a resistive electric circuit ($P = V.I$), the voltage and current have been measured and found to be $V = 100 \pm 2 \text{ V}$, $I = 10 \pm 0.2 \text{ A}$. Calculate the maximum possible error and the RSS uncertainty in the computation of the power, assuming the confidence levels for the uncertainties in V and I are the same.

- ❑ **Solution:**

- $\frac{\partial P}{\partial V} = I = 10 \text{ A}$, $\frac{\partial P}{\partial I} = V = 100 \text{ V}$.
- $(w_P)_{\max} = \left| \frac{\partial P}{\partial V} w_V \right| + \left| \frac{\partial P}{\partial I} w_I \right| = 10 \times 2 + 100 \times 0.2 = 40 \text{ W}$.
- $(w_P)_{\text{RSS}} = \left[\left(\frac{\partial P}{\partial V} w_V \right)^2 + \left(\frac{\partial P}{\partial I} w_I \right)^2 \right]^{1/2} = 28.3 \text{ W}$.

Example

Manometers are pressure-measuring devices that determine a pressure by measuring the height of a column of fluid. We would like to achieve an accuracy of 0.1% of the maximum reading, 10 kPa, by using a manometer whose uncertainty is 1/10 mm in reading the scale. Estimate the uncertainty that can be tolerated in the density of a gage fluid, which has a nominal value of 2500 kg/m³.

□ Solution:

- The relationship between the pressure and the measured column height for a manometer is: $P = \rho gh$, where ρ is the fluid density, g is the acceleration due to gravity, and h is the height of the column of fluid.
- The value of h for the maximum pressure reading is $h = \frac{P}{\rho g} = \frac{10000}{2500 \times 9.81} = 0.408$ m while the maximum uncertainty in pressure is 0.1% of 10 kPa, which is 10 Pa.
- We have: $w_P = \left[\left(\frac{\partial P}{\partial \rho} w_\rho \right)^2 + \left(\frac{\partial P}{\partial h} w_h \right)^2 \right]^{1/2} = \left[(ghw_\rho)^2 + (\rho g w_h)^2 \right]^{1/2} = 10$.
- Substituting the values for g , h , w_h in the above equation and solve for w_ρ , we have: $\left[(9.81 \times 0.408 \times w_\rho)^2 + (2500 \times 9.81 \times 0.0001)^2 \right]^{1/2} = 10 \rightarrow w_\rho = 2.4$.

Step-by-step procedure for the total uncertainty analysis

- ❑ The systematic and random uncertainties are combined to obtain the total uncertainty.

- For mean of x : $W_{\bar{x}} = (B_{\bar{x}}^2 + P_{\bar{x}}^2)^{1/2}$
- For single measurement of x : $W_x = (B_x^2 + P_x^2)^{1/2}$

- ❑ Step-by-step procedure for estimating the overall uncertainty

- 1. Define the measurement process:** involves reviewing test objectives, identifying all independent parameters and their nominal values, and defining the functional relationship between the independent parameters and the test results.
- 2. List all elemental error sources:** includes making a complete and exhaustive list of all possible error sources for each measured parameter.
 - To identify all uncertainties and maintain clear bookkeeping, group the uncertainties into categories based on their source: calibration, data acquisition, data reduction, methods, and others.

Step-by-step procedure for the total uncertainty analysis

3. Estimate the elemental errors: in this step, the systematic uncertainties and standard deviations must be estimated.

- If data are available to estimate the standard deviation of a parameter, or if the error is known to be random in nature, then it should be classified as random uncertainty; otherwise, classify it as systematic uncertainty.
- It is important to estimate all uncertainties to the same confidence level and for small samples, to identify the number of degrees of freedom associated with each standard deviation.
- The Table 7.1 can be used as a guideline.

TABLE 7.1 Guidelines for assigning elemental error

ERROR	ERROR TYPE
accuracy	systematic
common-mode volt	systematic
hysteresis	systematic
installation	systematic
linearity	systematic
loading	systematic
noise	random*
repeatability	random*
resolution/scale/ quantization	random*
spatial variation	systematic
thermal stability (gain, zero, etc.)	random*

*Assume that the number of samples is greater than 30 unless specified otherwise.

Step-by-step procedure for the total uncertainty analysis

4. Calculate the systematic and random uncertainties for each measured variable.

- Systematic uncertainties and standard deviations for the variable identified in Step 1 are calculated in this step.
- Data obtained in Step 3 are used in the calculations.

5. Propagate the systematic uncertainties and standard deviations all the way to the result(s).

- In this step, propagate the systematic and random uncertainties of the measured variables to the final test result(s).
- Care should be taken so that the **same confidence level** is observed in all calculations.

6. Calculate the total uncertainty of the results

- In this step, combine the systematic and random uncertainties to obtain the total uncertainty of the result(s).

Example: Transducer uncertainty

- ❑ Measuring pressure in chemical reaction tank using high quality pressure transducer. Manufacturing specifications of transducer:
 - Range: $\pm 3000\text{KPa}$ (3MPa).
 - Sensitivity: $\pm 0.25\%$ of FS.
 - Linearity: $\pm 0.15\%$ of FS.
 - Hysteresis: $\pm 0.10\%$ of FS.
- ❑ A large number of auxiliary tests been performed to check the repeatability of the pressure transducer at 1500KPa (working pressure), S_x from these test = 10KPa.
 - A large number of separate tests at the same pressure performed in data transmission system, $S_x = 5\text{KPa}$.
 - A/D converter – random uncertainty for 95% confidence level $P_x = 3\text{KPa}$. Assumed large sample size, $t = 2$ for 95% CL, $P_x = 2S_x$, $S_x = 3/2 = 1.5\text{KPa}$.
- ❑ Find the total uncertainty.

Example solution

- Find random uncertainty of the pressure measurement (single value): Calculating max. value.

- $S_x (\text{total}) = (10^2 + 5^2 + 1.5^2)^{1/2} = 11.3\text{KPa}.$
- $P_x = tS_x = 2.0 \cdot 11.3 = 22.6\text{KPa}.$ ($t=2$ for large sample size with 95% confidence level, $M=1$ when one measurement is taken).

- Systematic uncertainty of the pressure transducer:

Sensitivity = $(0.25 \cdot 3000)/100 = 7.5\text{KPa}.$

Linearity = $(0.15 \cdot 3000)/100 = 4.5\text{KPa}.$

Hysteresis = $(0.1 \cdot 3000)/100 = 3.0\text{KPa}$

- $B_x = (7.5^2 + 4.5^2 + 3.0^2)^{1/2} = 9.2\text{KPa}.$

- Total uncertainty at 95% confidence level:

- $W_x = (B_x^2 + P_x^2)^{1/2} = (22.6^2 + 9.2^2)^{1/2} = 22.8\text{KPa}$

- **(Maximum deviation from the true value)**

Example: Uncertainty in temperature sensor

- Find temperature sensor accuracy in a chemical process
- Systematic errors:
 - Calibration uncertainty - $\pm 0.5^{\circ}\text{C}$
 - Spatial variation - $\pm 2^{\circ}\text{C}$.
 - Installation effect - $\pm 1^{\circ}\text{C}$.
 - Linear relationship assumption pressure/voltage - $\pm 1^{\circ}\text{C}$.
- Random errors:
 - Sensor repeatability – 20 Measurements at the average value of 150°C , $S_{x_1} = 1.5^{\circ}\text{C}$.
 - Data transmission system test – 10 Measurements , $S_{x_2} = 0.5^{\circ}\text{C}$.

Example solution

- Total systematic uncertainty:
 - $B_x = (0.5^2 + 2^2 + 1^2 + 1^2)^{1/2} = 2.5^\circ\text{C}.$
- Random errors:
 - Combined $S_x = \left[(S_{x_1})^2 + (S_{x_2})^2 \right]^{1/2} = (1.5^2 + 0.5^2)^{1/2} = 1.58^\circ\text{C}.$
 - $\nu_1 = 20 - 1 = 19, \nu_2 = 10 - 1 = 9$
 - Combined $\nu = (1.5^2 + 0.5^2) / (1.5^4/19 + 0.5^4/9) = 22.9$, say 23.
 - $\nu = 23$, 95% confidence level $\rightarrow t = 2.07$ from t-distribution table.
- Random uncertainty of each temperature measurement ($M = 1$):
$$P_x = tS_x = 2.07 \times 1.58 = 3.3^\circ\text{C}.$$
- Total uncertainty of measurement with 95% CL:
$$W_x = (B_x^2 + P_x^2)^{1/2} = (2.5^2 + 3.3^2)^{1/2} = 4.14^\circ\text{C}.$$